Chapter 14

Truss Analysis Using the **Stiffness Method**





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Truss Analysis Using the **Stiffness Method**



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Truss Analysis Using the **Stiffness Method**

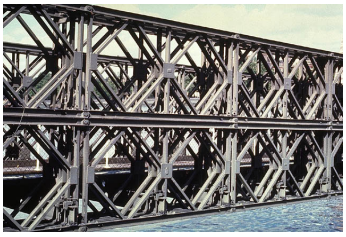


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Truss Analysis Using the **Stiffness Method**





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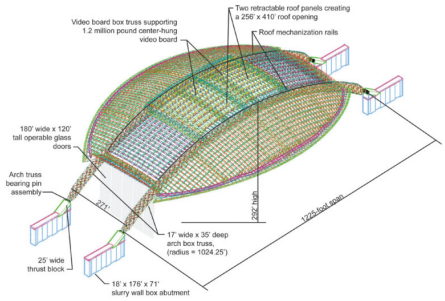
Truss Analysis Using the **Stiffness Method**



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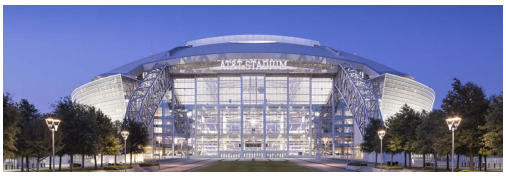
Truss Analysis Using the **Stiffness Method**



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Truss Analysis Using the **Stiffness Method**

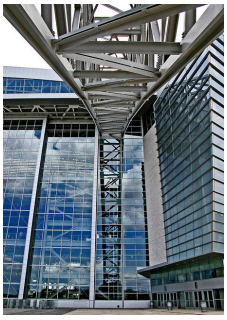


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Truss Analysis Using the **Stiffness Method**

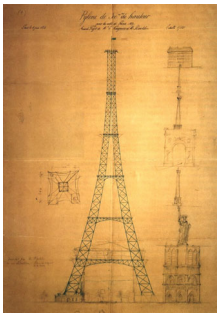


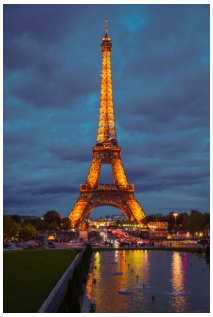


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Truss Analysis Using the **Stiffness Method**

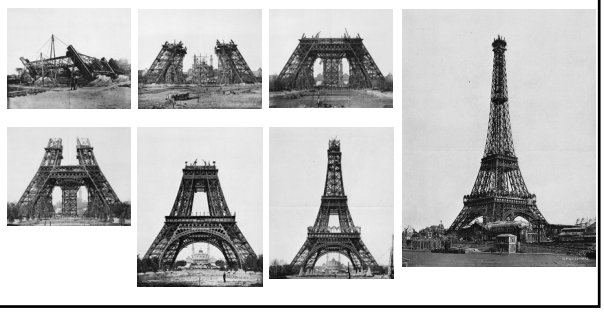




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Truss Analysis Using the **Stiffness Method**

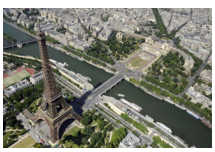


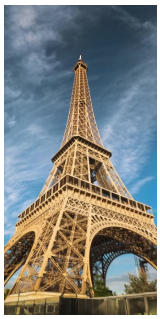
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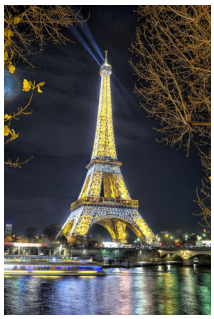
Chapter 14

Truss Analysis Using the **Stiffness Method**









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Stiffness method of analysis: **trusses**

Fundamentals of the Stiffness Method

- There are essentially two ways in which structures can be analyzed using matrix methods.
- The ***stiffness method***, to be used in this and the following chapters, is a **displacement method**.
- A **force method**, called the ***flexibility method***, as outlined in Sec. 9.2, can also be used; however, this method will not be presented here.

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Stiffness method of analysis: **trusses**

Fundamentals of the Stiffness Method

- There are several reasons for this.
- Most importantly, the stiffness method can be used to analyze both ***statically determinate*** and ***statically indeterminate*** structures.
- The flexibility method requires a different procedure for each of these two cases.
- Also, it is generally easier to formulate the necessary matrices for computer operations using the ***stiffness method***; once this is done, the calculations can be performed efficiently.

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Stiffness method of analysis: **trusses**

Fundamentals of the Stiffness Method

- Application of the stiffness method requires subdividing the structure into a series of discrete ***finite elements*** and then identifying their endpoints as ***nodes***.
- For truss analysis, the ***finite elements*** are represented by each of the members of the truss, and the ***nodes*** represent the joints.
- The ***force-displacement*** relationships for each element are determined.
- Then, these relationships are related to one another using the **force equilibrium equations** written at the ***nodes***.

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Stiffness method of analysis: **trusses**

Fundamentals of the Stiffness Method

- These relationships, for the entire structure, are then grouped together into what is called the ***structure stiffness matrix*** **K**.
- Once it is established, the ***unknown displacements*** of the nodes can then be determined for any given loading on the truss.
- When these ***displacements*** are known, the external and internal forces in the truss can be calculated using the ***force-displacement relations*** for each member.

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Stiffness method of analysis: **trusses**

Member and Node Identification

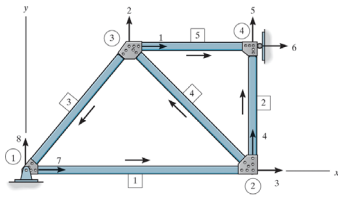
- One of the first steps when applying the stiffness method is to identify the elements or members of the truss and their nodes.
- Each ***member*** is identified by a number within a ***square*** □.
- Each ***joint*** is identified by a number within a ***circle*** ○.
- Also, the "***near***" and "***far***" ends of the member must be identified.
- This will be done using an ***arrow*** written along the member, with the head of the arrow ***directed toward the far end***.

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Stiffness method of analysis: **trusses**

Member and Node Identification

- Examples of member, ***node***, and "***direction***" identification for a truss are shown below.
- These assignments have all been done ***arbitrarily***.*



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Stiffness method of analysis: **trusses**

Global and Member Coordinates

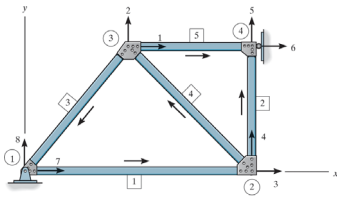
- Since loads and displacements are ***vector quantities***, it is necessary to establish a ***coordinate system*** to specify their sense of direction.
- Here we will use ***two different types of coordinate systems***.

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Stiffness method of analysis: **trusses**

Global and Member Coordinates

- A single ***global*** or ***structure coordinate system***, *x*, *y*, will be used to specify the sense of each of the ***external*** force and displacement components at the nodes.

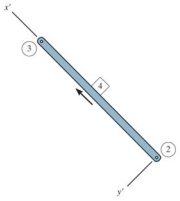


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Stiffness method of analysis: **trusses**

Global and Member Coordinates

- A ***local*** or ***member coordinate system*** will be used for each member to specify the sense of direction of its ***displacements*** and ***internal loadings***.
- This system will be identified using ***axes*** with the origin at the "***near***" node and the axis extending toward the "***far***" node.



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Stiffness method of analysis: **trusses**

Kinematic Indeterminacy

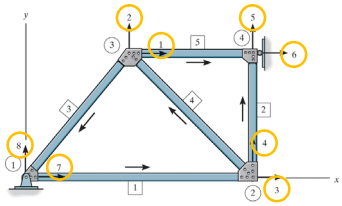
- The ***unconstrained displacements*** for a truss joint represent the primary ***unknowns*** of any displacement method, and therefore these must be identified.
- There are ***two degrees of freedom***, or two displacements, for each joint (***node***).
- For application, each degree of freedom will be specified on the truss using a ***code number***.
- The ***code number*** is shown at the joint or node and referenced to its ***positive global coordinate*** direction.

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Stiffness method of analysis: **trusses**

Kinematic Indeterminacy

- For example, the truss below has ***eight degrees of freedom***, which have been identified by the ***code numbers*** 1 through 8.

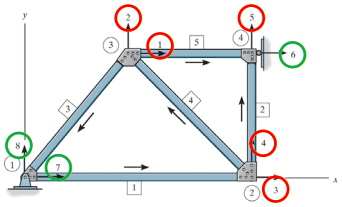


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Stiffness method of analysis: **trusses**

Kinematic Indeterminacy

- Of the eight possible displacements 1 through 5 represent unknown or ***unconstrained degrees of freedom***, and 6 through 8 represent ***constrained degrees of freedom***.
- The truss is ***kinematically indeterminate to the fifth degree.***



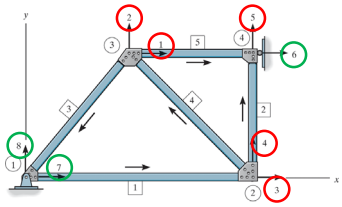
- Due to the pin and roller, these displacements are zero

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Stiffness method of analysis: **trusses**

Kinematic Indeterminacy

- In this course, *the* ***lowest code numbers*** *will be used to identify the unknown displacements (unconstrained degrees of freedom)*
- The ***highest code numbers*** *will be used to identify the known displacements (constrained degrees of freedom).*

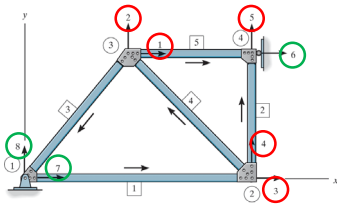


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Stiffness method of analysis: **trusses**

Kinematic Indeterminacy

- This method of identification is convenience when partitioning the structure stiffness matrix **K** so that the ***unknown displacements*** can be found in the most direct manner.

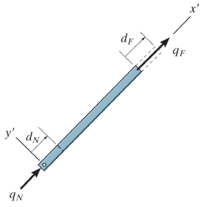


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Stiffness method of analysis: **trusses**

Member Stiffness Matrix

- We will establish the ***stiffness matrix for a single truss member*** using ***local coordinates***, oriented as shown below.
- The terms in this matrix represent the ***load-displacement*** relations for the member.



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Stiffness method of analysis: **trusses**

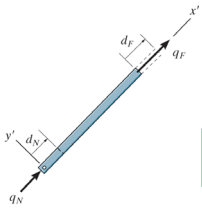
Member Stiffness Matrix

- Since the loads on a truss member ***only act along the member***, then the displacements of the ***nodes*** are only along the axis.
- To obtain the load-displacement relations, we will apply two independent displacements to the member.

$$\sigma = E\varepsilon$$

$$\frac{P}{A} = E\left(\frac{\delta}{L}\right)$$

$$P = \frac{EA}{L}\delta$$



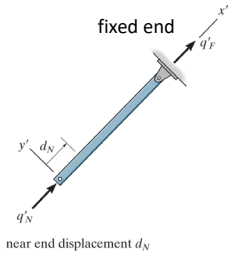
$$q_i = \frac{EA}{L}d_i$$

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Stiffness method of analysis: **trusses**

Member Stiffness Matrix

- When a ***positive displacement*** occurs on the ***near*** end of the member, while the far end is held pinned (fixed), the forces developed at the ends of the member are:



near end displacement d_N

$$q'_N = \frac{AE}{L}d_N$$

$$q'_F = -\frac{AE}{L}d_N$$

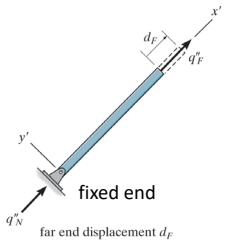
- Here $\boldsymbol{q'_F}$ is negative since for equilibrium it must act in the negative direction.

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Stiffness method of analysis: **trusses**

Member Stiffness Matrix

- Likewise, a positive displacement at the ***far*** end, keeping the near end pinned (fixed)



far end displacement d_F

$$q''_F = \frac{AE}{L}d_F$$

$$q''_N = -\frac{AE}{L}d_F$$

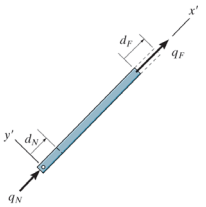
- Here $\boldsymbol{q''_N}$ is negative since for equilibrium it must act in the negative direction.

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Stiffness method of analysis: **trusses**

Member Stiffness Matrix

- By superposition, the resultant forces caused by both displacements are



near and far end displacements d_N and d_F

$$q_N = q'_N + q''_N$$

$$q_N = \frac{AE}{L}d_N - \frac{AE}{L}d_F$$

$$q_F = q'_F + q''_F$$

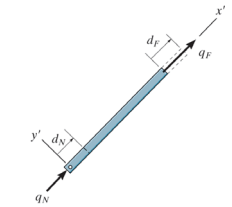
$$q_F = -\frac{AE}{L}d_N + \frac{AE}{L}d_F$$

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Stiffness method of analysis: **trusses**

Member Stiffness Matrix

- These load-displacement equations may be written in ***matrix form*** as:



near and far end displacements d_N and d_F

$$\begin{Bmatrix} q_N \\ q_F \end{Bmatrix} = \frac{AE}{L}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}\begin{Bmatrix} d_N \\ d_F \end{Bmatrix}$$

$$\mathbf{q} = \mathbf{k}'\mathbf{d}$$

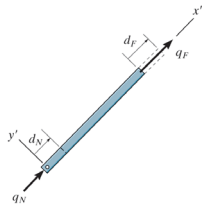
$$\mathbf{k}' = \frac{AE}{L}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

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Stiffness method of analysis: **trusses**

Member Stiffness Matrix

- Please review **Appendix A** in the textbook for an introduction to matrix algebra.



near and far end displacements d_N and d_F

$$\begin{Bmatrix} q_N \\ q_F \end{Bmatrix} = \frac{AE}{L}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}\begin{Bmatrix} d_N \\ d_F \end{Bmatrix}$$

$$\mathbf{q} = \mathbf{k}'\mathbf{d}$$

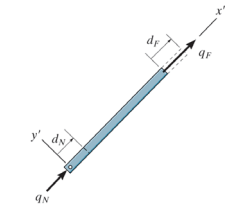
$$\mathbf{k}' = \frac{AE}{L}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

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Stiffness method of analysis: **trusses**

Member Stiffness Matrix

- This matrix **k'** is called the ***member stiffness matrix***, and it has the same form for each member of the truss.



near and far end displacements d_N and d_F

$$\begin{Bmatrix} q_N \\ q_F \end{Bmatrix} = \frac{AE}{L}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}\begin{Bmatrix} d_N \\ d_F \end{Bmatrix}$$

$$\mathbf{q} = \mathbf{k}'\mathbf{d}$$

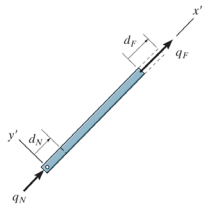
$$\mathbf{k}' = \frac{AE}{L}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

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Stiffness method of analysis: **trusses**

Member Stiffness Matrix

- The four elements that comprise it are called ***member stiffness influence coefficients***, $\mathbf{k}'_{ij}$.



near and far end displacements d_N and d_F

$$\begin{Bmatrix} q_N \\ q_F \end{Bmatrix} = \frac{AE}{L}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}\begin{Bmatrix} d_N \\ d_F \end{Bmatrix}$$

$$\mathbf{q} = \mathbf{k}'\mathbf{d}$$

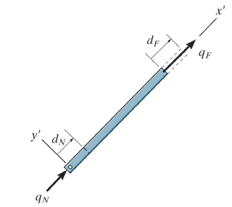
$$\mathbf{k}' = \frac{AE}{L}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

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Stiffness method of analysis: **trusses**

Member Stiffness Matrix

- Physically, $\mathbf{k}'_{ij}$ represents the force at joint *i* when a ***unit displacement*** is imposed at joint *j*.



near and far end displacements d_N and d_F

$$\begin{Bmatrix} q_N \\ q_F \end{Bmatrix} = \frac{AE}{L}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}\begin{Bmatrix} d_N \\ d_F \end{Bmatrix}$$

$$\mathbf{q} = \mathbf{k}'\mathbf{d}$$

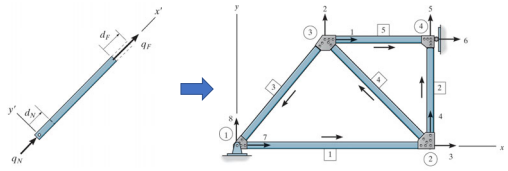
$$\mathbf{k}' = \frac{AE}{L}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

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Stiffness method of analysis: **trusses**

Displacement and Force Transformation Matrices

- Since a truss is composed of many members (elements), we must now develop a method for transforming the member forces **q** and displacements **d** defined in local coordinates to global coordinates.



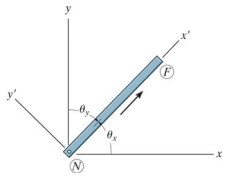
Local coordinates Global coordinates

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Stiffness method of analysis: **trusses**

Displacement and Force Transformation Matrices

- To do this, we will define the direction of each member using the ***smallest angles*** between the ***positive x and y*** global axes and the ***positive x'*** local axis.
- These angles are θ_x and θ_y as shown below:

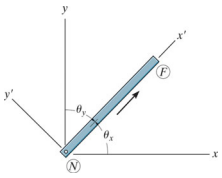


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Stiffness method of analysis: **trusses**

Displacement and Force Transformation Matrices

- The cosines of these angles will be used in the matrix analysis that follows, where $\lambda_x = \cos\theta_x$ and $\lambda_y = \cos\theta_y$
- Numerical values for λ_x and λ_y can easily be generated by a computer once the *x*, *y* coordinates of the near end ***N*** and far end ***F*** of the member have been specified.

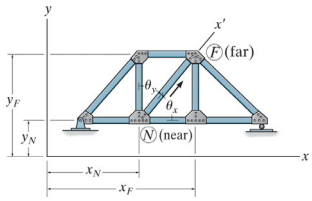


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Stiffness method of analysis: **trusses**

Displacement and Force Transformation Matrices

- For example, consider member *NF* of the truss shown below.



$$\lambda_x = \cos\theta_x = \frac{x_F - x_N}{L_{NF}} = \frac{x_F - x_N}{\sqrt{(x_F - x_N)^2 + (y_F - y_N)^2}}$$

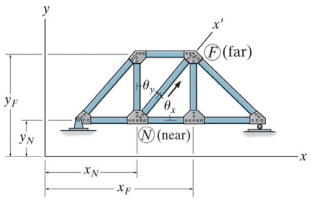
$$\lambda_y = \cos\theta_y = \frac{y_F - y_N}{L_{NF}} = \frac{y_F - y_N}{\sqrt{(x_F - x_N)^2 + (y_F - y_N)^2}}$$

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Stiffness method of analysis: **trusses**

Displacement and Force Transformation Matrices

- The algebraic signs in these ***"generalized"*** equations will automatically account for members that are oriented in other quadrants of the *x*–*y* plane.



$$\lambda_x = \cos\theta_x = \frac{x_F - x_N}{L_{NF}} = \frac{x_F - x_N}{\sqrt{(x_F - x_N)^2 + (y_F - y_N)^2}}$$

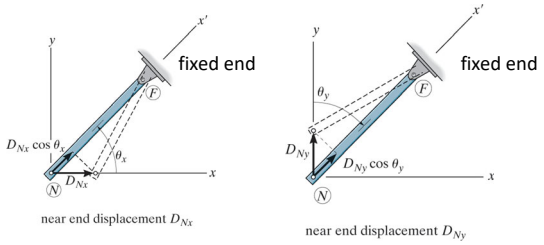
$$\lambda_y = \cos\theta_y = \frac{y_F - y_N}{L_{NF}} = \frac{y_F - y_N}{\sqrt{(x_F - x_N)^2 + (y_F - y_N)^2}}$$

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Stiffness method of analysis: **trusses**

Displacement Transformation Matrices

- In global coordinates, each end of the member can have ***two degrees of freedom*** or independent displacements; namely, joint ***N*** has D_{Nx} and D_{Ny}.



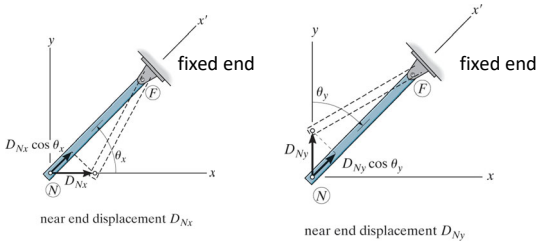
41

Stiffness method of analysis: **trusses**

Displacement Transformation Matrices

- The effect of *both* global displacements causes the local displacement:

$$d_N = D_{Nx}\cos\theta_x + D_{Ny}\cos\theta_y$$

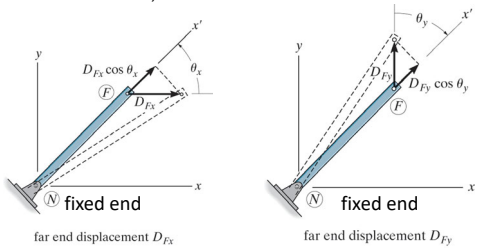


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Stiffness method of analysis: **trusses**

Displacement Transformation Matrices

- In global coordinates, each end of the member can have two degrees of freedom or independent displacements; namely, joint ***F*** has D_{Fx} and D_{Fy}.



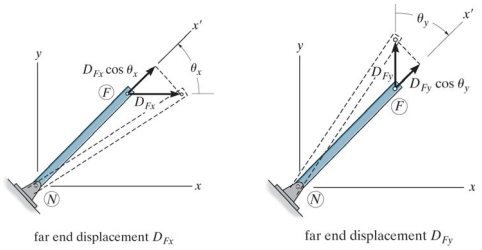
43

Stiffness method of analysis: **trusses**

Displacement Transformation Matrices

- The effect of *both* global displacements causes the local displacement:

$$d_F = D_{Fx}\cos\theta_x + D_{Fy}\cos\theta_y$$



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Stiffness method of analysis: **trusses**

Displacement Transformation Matrices

- Since: $\lambda_x = \cos\theta_x \qquad \lambda_y = \cos\theta_y$

$$d_N = D_{Nx}\cos\theta_x + D_{Ny}\cos\theta_y = D_{Nx}\lambda_x + D_{Ny}\lambda_y$$

$$d_F = D_{Fx}\cos\theta_x + D_{Fy}\cos\theta_y = D_{Fx}\lambda_x + D_{Fy}\lambda_y$$

$$\underset{\mathbf{d}}{\begin{Bmatrix} d_N \\ d_F \end{Bmatrix}} = \underset{\mathbf{T}}{\begin{bmatrix} \lambda_x & \lambda_y & 0 & 0 \\ 0 & 0 & \lambda_x & \lambda_y \end{bmatrix}} \underset{\mathbf{D}}{\begin{Bmatrix} D_{Nx} \\ D_{Ny} \\ D_{Fx} \\ D_{Fy} \end{Bmatrix}}$$

$$\mathbf{d} = \mathbf{TD}$$

$$\mathbf{T} = \begin{bmatrix} \lambda_x & \lambda_y & 0 & 0 \\ 0 & 0 & \lambda_x & \lambda_y \end{bmatrix}$$

- **T** transforms the four global *x*, *y* displacements **D** into the two local displacements **d**.

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Stiffness method of analysis: **trusses**

Displacement Transformation Matrices

- Since: $\lambda_x = \cos\theta_x \qquad \lambda_y = \cos\theta_y$

$$d_N = D_{Nx}\cos\theta_x + D_{Ny}\cos\theta_y = D_{Nx}\lambda_x + D_{Ny}\lambda_y$$

$$d_F = D_{Fx}\cos\theta_x + D_{Fy}\cos\theta_y = D_{Fx}\lambda_x + D_{Fy}\lambda_y$$

$$\underset{\mathbf{d}}{\begin{Bmatrix} d_N \\ d_F \end{Bmatrix}} = \underset{\mathbf{T}}{\begin{bmatrix} \lambda_x & \lambda_y & 0 & 0 \\ 0 & 0 & \lambda_x & \lambda_y \end{bmatrix}} \underset{\mathbf{D}}{\begin{Bmatrix} D_{Nx} \\ D_{Ny} \\ D_{Fx} \\ D_{Fy} \end{Bmatrix}}$$

$$\mathbf{d} = \mathbf{TD}$$

$$\mathbf{T} = \begin{bmatrix} \lambda_x & \lambda_y & 0 & 0 \\ 0 & 0 & \lambda_x & \lambda_y \end{bmatrix}$$

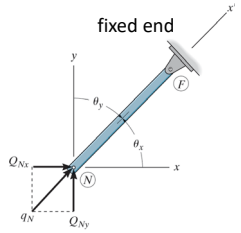
- Hence, **T** is referred to as the ***displacement transformation matrix***.

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Stiffness method of analysis: **trusses**

Force Transformation Matrices

- Consider now the application of the force q_N to the ***near end*** of the member, with the far end held pinned (fixed).



$$Q_{Nx} = q_N\cos\theta_x$$

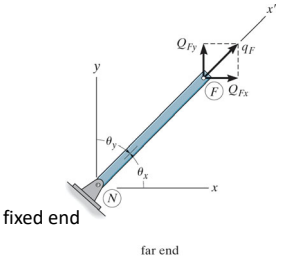
$$Q_{Ny} = q_N\cos\theta_y$$

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Stiffness method of analysis: **trusses**

Force Transformation Matrices

- Likewise, if the force q_F to the ***far end*** of the member, with the near end held pinned (fixed).



$$Q_{Fx} = q_F\cos\theta_x$$

$$Q_{Fy} = q_F\cos\theta_y$$

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Stiffness method of analysis: **trusses**

Force Transformation Matrices

- Since: $\lambda_x = \cos\theta_x$ $\lambda_y = \cos\theta_y$

$$Q_{Nx} = q_N\lambda_x \quad Q_{Ny} = q_N\lambda_y \quad Q_{Fx} = q_F\lambda_x \quad Q_{Fy} = q_F\lambda_y$$

$$\begin{Bmatrix} Q_{Nx} \\ Q_{Ny} \\ Q_{Fx} \\ Q_{Fy} \end{Bmatrix} = \begin{bmatrix} \lambda_x & 0 \\ \lambda_y & 0 \\ 0 & \lambda_x \\ 0 & \lambda_y \end{bmatrix} \begin{Bmatrix} q_N \\ q_F \end{Bmatrix} \qquad \mathbf{T} = \begin{bmatrix} \lambda_x & \lambda_y & 0 & 0 \\ 0 & 0 & \lambda_x & \lambda_y \end{bmatrix}$$

$$\mathbf{T}^T = \begin{bmatrix} \lambda_x & 0 \\ \lambda_y & 0 \\ 0 & \lambda_x \\ 0 & \lambda_y \end{bmatrix} \qquad \mathbf{Q} = \mathbf{T}^T\mathbf{q}$$

- In this case this ***force transformation matrix*** transforms the two local (*x'*) forces **q** acting at the ends of the member into the four global (*x*, *y*) force components **Q**.

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Stiffness method of analysis: **trusses**

Force Transformation Matrices

- Since: $\lambda_x = \cos\theta_x$ $\lambda_y = \cos\theta_y$

$$Q_{Nx} = q_N\lambda_x \quad Q_{Ny} = q_N\lambda_y \quad Q_{Fx} = q_F\lambda_x \quad Q_{Fy} = q_F\lambda_y$$

$$\begin{Bmatrix} Q_{Nx} \\ Q_{Ny} \\ Q_{Fx} \\ Q_{Fy} \end{Bmatrix} = \begin{bmatrix} \lambda_x & 0 \\ \lambda_y & 0 \\ 0 & \lambda_x \\ 0 & \lambda_y \end{bmatrix} \begin{Bmatrix} q_N \\ q_F \end{Bmatrix} \qquad \mathbf{T} = \begin{bmatrix} \lambda_x & \lambda_y & 0 & 0 \\ 0 & 0 & \lambda_x & \lambda_y \end{bmatrix}$$

$$\mathbf{T}^T = \begin{bmatrix} \lambda_x & 0 \\ \lambda_y & 0 \\ 0 & \lambda_x \\ 0 & \lambda_y \end{bmatrix} \qquad \mathbf{Q} = \mathbf{T}^T\mathbf{q}$$

- Notice that $\mathbf{T}^T$ is the transpose of the displacement transformation matrix **T**.

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Stiffness method of analysis: **trusses**

Member Global Stiffness Matrix

- We will now combine the results of the preceding sections and thereby determine the stiffness matrix for a member in terms of its global force components **Q** and its global displacements **D**.
- Substitute **d** = **TD** into **q** = **k'd** so that the member's forces **q** are then expressed in terms of the global displacements **D**.

$$\mathbf{q} = \mathbf{k'd} \quad \Rightarrow \quad \mathbf{q} = \mathbf{k'TD}$$

- Substituting this equation into $\mathbf{Q} = \mathbf{T}^T\mathbf{q}$ yields

$$\mathbf{Q} = \mathbf{T}^T\mathbf{q} \;\Rightarrow\; \mathbf{Q} = \mathbf{T}^T\mathbf{k'TD} \qquad \boxed{\mathbf{k} = \mathbf{T}^T\mathbf{k'T}}$$

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Stiffness method of analysis: **trusses**

Member Global Stiffness Matrix

- The matrix **k** is the ***member stiffness matrix*** in global coordinates.
- Since $\mathbf{T}^T$, **T** and **k'** are known, we have

$$\mathbf{k} = \underbrace{\begin{bmatrix} \lambda_x & 0 \\ \lambda_y & 0 \\ 0 & \lambda_x \\ 0 & \lambda_y \end{bmatrix}_{4x2}}_{\mathbf{T}'} \underbrace{\frac{AE}{L}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}_{2x2}}_{\mathbf{k}'} \underbrace{\begin{bmatrix} \lambda_x & \lambda_y & 0 & 0 \\ 0 & 0 & \lambda_x & \lambda_y \end{bmatrix}_{2x4}}_{\mathbf{T}}$$

$$\mathbf{k} = \frac{AE}{L}\begin{bmatrix} \lambda_x & 0 \\ \lambda_y & 0 \\ 0 & \lambda_x \\ 0 & \lambda_y \end{bmatrix}_{4x2} \begin{bmatrix} \lambda_x & \lambda_y & -\lambda_x & -\lambda_y \\ -\lambda_x & -\lambda_y & \lambda_x & \lambda_y \end{bmatrix}_{2x4}$$

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Stiffness method of analysis: **trusses**

Member Global Stiffness Matrix

- The matrix **k** is the ***member stiffness matrix*** in global coordinates.
- Since $\mathbf{T}^T$, **T** and **k'** are known, we have

$$\mathbf{k} = \frac{AE}{L}\begin{array}{c} \\ N_x \\ N_y \\ F_x \\ F_y \end{array}\begin{array}{c} \begin{array}{cccc} N_x & N_y & F_x & F_y \end{array} \\ \begin{bmatrix} \lambda_x^2 & \lambda_x\lambda_y & -\lambda_x^2 & -\lambda_x\lambda_y \\ \lambda_x\lambda_y & \lambda_y^2 & -\lambda_x\lambda_y & -\lambda_y^2 \\ -\lambda_x^2 & -\lambda_x\lambda_y & \lambda_x^2 & \lambda_x\lambda_y \\ -\lambda_x\lambda_y & -\lambda_y^2 & \lambda_x\lambda_y & \lambda_y^2 \end{bmatrix} \end{array}$$

Member stiffness matrix in global coordinates

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Stiffness method of analysis: **trusses**

Member Global Stiffness Matrix

- This is indicated by the code number notation along the rows and columns, that is, N_x , N_y , F_x , and F_y .

$$\mathbf{k} = \frac{AE}{L}\begin{array}{c} \\ N_x \\ N_y \\ F_x \\ F_y \end{array}\begin{array}{c} \begin{array}{cccc} N_x & N_y & F_x & F_y \end{array} \\ \begin{bmatrix} \lambda_x^2 & \lambda_x\lambda_y & -\lambda_x^2 & -\lambda_x\lambda_y \\ \lambda_x\lambda_y & \lambda_y^2 & -\lambda_x\lambda_y & -\lambda_y^2 \\ -\lambda_x^2 & -\lambda_x\lambda_y & \lambda_x^2 & \lambda_x\lambda_y \\ -\lambda_x\lambda_y & -\lambda_y^2 & \lambda_x\lambda_y & \lambda_y^2 \end{bmatrix} \end{array}$$

Member stiffness matrix in global coordinates

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Stiffness method of analysis: **trusses**

Member Global Stiffness Matrix

- The ***location*** of each element in this symmetric matrix is referenced with each global degree of freedom associated with the ***near end N***, followed by the ***far end F***.

$$\mathbf{k} = \frac{AE}{L}\begin{bmatrix} \lambda_x^2 & \lambda_x\lambda_y & -\lambda_x^2 & -\lambda_x\lambda_y \\ \lambda_x\lambda_y & \lambda_y^2 & -\lambda_x\lambda_y & -\lambda_y^2 \\ -\lambda_x^2 & -\lambda_x\lambda_y & \lambda_x^2 & \lambda_x\lambda_y \\ -\lambda_x\lambda_y & -\lambda_y^2 & \lambda_x\lambda_y & \lambda_y^2 \end{bmatrix} \begin{matrix} N_x \\ N_y \\ F_x \\ F_y \end{matrix} \quad \begin{matrix} N_x & N_y & F_x & F_y \end{matrix}$$

Member stiffness matrix in global coordinates

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Stiffness method of analysis: **trusses**

Member Global Stiffness Matrix

- Here **k** represents the ***force-displacement*** relations for the member when the components of force and displacement at the ends of the member are in the global or *x*, *y* directions.

$$\mathbf{k} = \frac{AE}{L}\begin{bmatrix} \lambda_x^2 & \lambda_x\lambda_y & -\lambda_x^2 & -\lambda_x\lambda_y \\ \lambda_x\lambda_y & \lambda_y^2 & -\lambda_x\lambda_y & -\lambda_y^2 \\ -\lambda_x^2 & -\lambda_x\lambda_y & \lambda_x^2 & \lambda_x\lambda_y \\ -\lambda_x\lambda_y & -\lambda_y^2 & \lambda_x\lambda_y & \lambda_y^2 \end{bmatrix} \begin{matrix} N_x \\ N_y \\ F_x \\ F_y \end{matrix} \quad \begin{matrix} N_x & N_y & F_x & F_y \end{matrix}$$

Member stiffness matrix in global coordinates

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Stiffness method of analysis: **trusses**

Member Global Stiffness Matrix

- The **k** matrix is symmetric.
- The **k** matrix is made up of three values λ_x^2, λ_y^2, and λ_{xy}

$$\mathbf{k} = \frac{AE}{L}\begin{bmatrix} \lambda_x^2 & \lambda_x\lambda_y & -\lambda_x^2 & -\lambda_x\lambda_y \\ \lambda_x\lambda_y & \lambda_y^2 & -\lambda_x\lambda_y & -\lambda_y^2 \\ -\lambda_x^2 & -\lambda_x\lambda_y & \lambda_x^2 & \lambda_x\lambda_y \\ -\lambda_x\lambda_y & -\lambda_y^2 & \lambda_x\lambda_y & \lambda_y^2 \end{bmatrix} \begin{matrix} N_x \\ N_y \\ F_x \\ F_y \end{matrix} \quad \begin{matrix} N_x & N_y & F_x & F_y \end{matrix}$$

Member stiffness matrix in global coordinates

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Stiffness method of analysis: **trusses**

Truss Stiffness Matrix

- The member stiffness matrices in global coordinates are then assembled to build the ***structure stiffness matrix*** **K**.
- This process of combining the ***member stiffness*** matrices depends on careful identification of the row and column elements.
- The assembly is based on the *four* code numbers:
  N_x , N_y , F_x , and F_y

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Stiffness method of analysis: **trusses**

Truss Stiffness Matrix

- For all the truss members, the structure stiffness matrix will then have an order equal to the ***highest code number.***
- This is the ***total number of degrees of freedom for the truss.***
- To form the structure stiffness matrix, **K**, it is necessary to take each member's elements in **k** and place them in the ***same row and column*** designation in **K**.
- If some elements are assigned to the same location, then they must be ***added together algebraically***.

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Stiffness method of analysis: **trusses**

Truss Stiffness Matrix

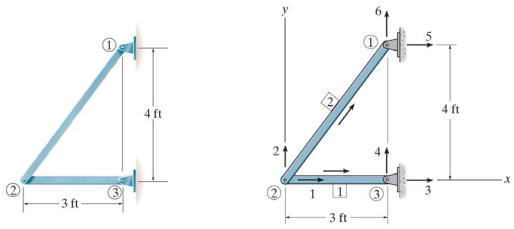
- This method of assembling the member stiffness matrices to form the structure stiffness matrix will now be demonstrated by two numerical examples.
- Although this process is ***somewhat tedious when done by hand***, it is rather easy to program on a computer.

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Stiffness method of analysis: **trusses**

Example 14.1

- Determine the structure stiffness matrix for the two-member truss shown below. *AE* is constant.

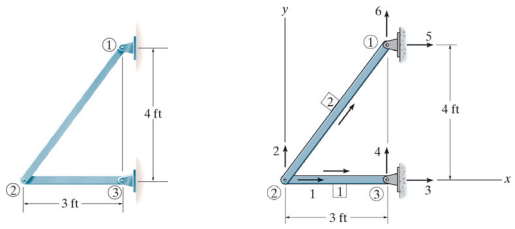


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Stiffness method of analysis: **trusses**

Example 14.1

- By inspection, joint ② will have two unknown displacement components, whereas joints ① and ③ are constrained from displacement.

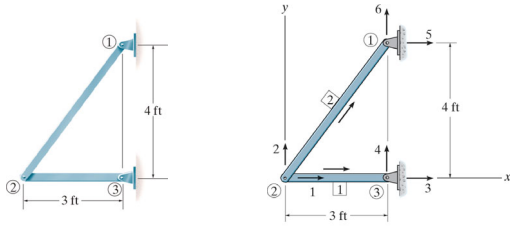


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Stiffness method of analysis: **trusses**

Example 14.1

- Consequently, the displacements at joint ② are code numbered first, followed by those at joints ③ and, ①

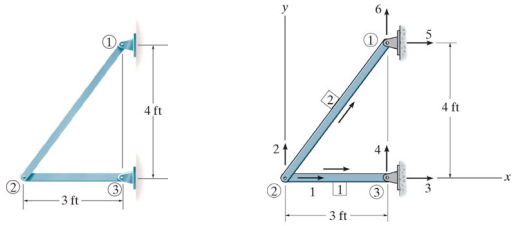


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Stiffness method of analysis: **trusses**

Example 14.1

- The origin of the global coordinate system is at joint ②.

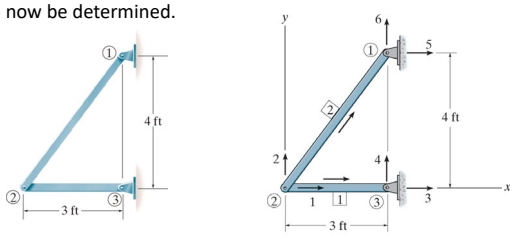


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Stiffness method of analysis: **trusses**

Example 14.1

- The members are identified, and arrows are written along the two members to identify the near and far ends of each member.
- The direction cosines and the stiffness matrix for each member can now be determined.



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Stiffness method of analysis: **trusses**

Example 14.1 – Member 1

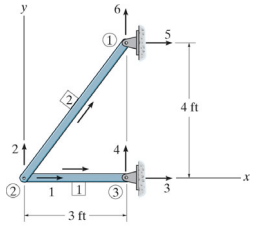
- Since ② is the near end and ③ is the far end, then:

$$\lambda_x = \frac{x_F - x_N}{L_{NF}} = \frac{3-0}{3} = 1$$

$$\lambda_y = \frac{y_F - y_N}{L_{NF}} = \frac{0-0}{3} = 0$$

$$\mathbf{k}_1 = AE \begin{bmatrix} 0.333 & 0 & -0.333 & 0 \\ 0 & 0 & 0 & 0 \\ -0.333 & 0 & 0.333 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix}$$

$$\mathbf{k} = \frac{AE}{L} \begin{bmatrix} \lambda_x^2 & \lambda_x\lambda_y & -\lambda_x^2 & -\lambda_x\lambda_y \\ \lambda_x\lambda_y & \lambda_y^2 & -\lambda_x\lambda_y & -\lambda_y^2 \\ -\lambda_x^2 & -\lambda_x\lambda_y & \lambda_x^2 & \lambda_x\lambda_y \\ -\lambda_x\lambda_y & -\lambda_y^2 & \lambda_x\lambda_y & \lambda_y^2 \end{bmatrix} \begin{matrix} N_x \\ N_y \\ F_x \\ F_y \end{matrix}$$



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Stiffness method of analysis: **trusses**

Example 14.1 – Member 2

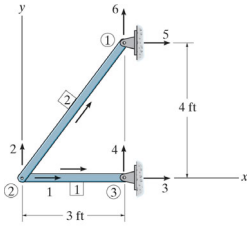
- Since ② is the near end and ① is the far end, we have:

$$\lambda_x = \frac{x_F - x_N}{L_{NF}} = \frac{3-0}{5} = 0.6$$

$$\lambda_y = \frac{y_F - y_N}{L_{NF}} = \frac{4-0}{5} = 0.8$$

$$\mathbf{k} = \frac{AE}{L}\begin{bmatrix} \lambda_x^2 & \lambda_x\lambda_y & -\lambda_x^2 & -\lambda_x\lambda_y \\ \lambda_x\lambda_y & \lambda_y^2 & -\lambda_x\lambda_y & -\lambda_y^2 \\ -\lambda_x^2 & -\lambda_x\lambda_y & \lambda_x^2 & \lambda_x\lambda_y \\ -\lambda_x\lambda_y & -\lambda_y^2 & \lambda_x\lambda_y & \lambda_y^2 \end{bmatrix} \begin{matrix} N_x \\ N_y \\ F_x \\ F_y \end{matrix}$$

$$\mathbf{k}_2 = AE\begin{bmatrix} 0.072 & 0.096 & -0.072 & -0.096 \\ 0.096 & 0.128 & -0.096 & -0.128 \\ -0.072 & -0.096 & 0.072 & 0.096 \\ -0.096 & -0.128 & 0.096 & 0.128 \end{bmatrix} \begin{matrix} 1 \\ 2 \\ 5 \\ 6 \end{matrix} \quad \begin{matrix} 1 & 2 & 5 & 6 \end{matrix}$$

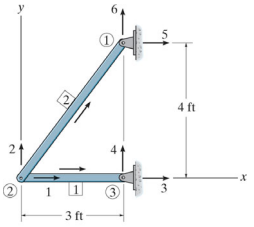


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Stiffness method of analysis: **trusses**

Example 14.1 – Structure Stiffness Matrix

- This matrix has an order of 6 × 6 since there are ***six designated degrees of freedom*** for the truss.



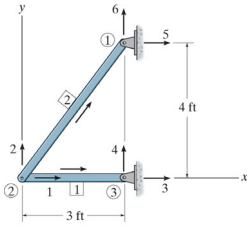
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Stiffness method of analysis: **trusses**

Example 14.1 – Structure Stiffness Matrix

- Corresponding elements of the member stiffness matrices are now added algebraically to form the structure stiffness matrix.

$$\mathbf{K} = \mathbf{k}_1 + \mathbf{k}_2$$



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Stiffness method of analysis: **trusses**

Example 14.1 – Structure Stiffness Matrix

- The assembly process is easier to see if the missing numerical columns and rows in $\mathbf{k}_1$ and $\mathbf{k}_2$ are ***filled with zeros*** to form two matrices.

$$\mathbf{K} = \mathbf{k}_1 + \mathbf{k}_2$$

$$\mathbf{k}_1 = AE\begin{bmatrix} 0.333 & 0 & -0.333 & 0 \\ 0 & 0 & 0 & 0 \\ -0.333 & 0 & 0.333 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} \quad \begin{matrix} 1 & 2 & 3 & 4 \end{matrix}$$

$$= AE\begin{bmatrix} 0.333 & 0 & -0.333 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -0.333 & 0 & 0.333 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{matrix} \quad \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 \end{matrix}$$

$$\mathbf{k}_2 = AE\begin{bmatrix} 0.072 & 0.096 & -0.072 & -0.096 \\ 0.096 & 0.128 & -0.096 & -0.128 \\ -0.072 & -0.096 & 0.072 & 0.096 \\ -0.096 & -0.128 & 0.096 & 0.128 \end{bmatrix} \begin{matrix} 1 \\ 2 \\ 5 \\ 6 \end{matrix} \quad \begin{matrix} 1 & 2 & 5 & 6 \end{matrix}$$

$$= AE\begin{bmatrix} 0.072 & 0.096 & 0 & 0 & -0.072 & -0.096 \\ 0.096 & 0.128 & 0 & 0 & -0.096 & -0.128 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -0.072 & -0.096 & 0 & 0 & 0.072 & 0.096 \\ -0.096 & -0.128 & 0 & 0 & 0.096 & 0.128 \end{bmatrix} \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{matrix} \quad \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 \end{matrix}$$

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Stiffness method of analysis: **trusses**

Example 14.1 – Structure Stiffness Matrix

- The assembly process is easier to see if the missing numerical columns and rows in $\mathbf{k}_1$ and $\mathbf{k}_2$ are ***filled with zeros*** to form two matrices.

$$\mathbf{K} = \mathbf{k}_1 + \mathbf{k}_2$$

$$\mathbf{K} = AE\begin{bmatrix} 0.333 & 0 & -0.333 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -0.333 & 0 & 0.333 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{matrix} \quad \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 \end{matrix}$$

$$+ AE\begin{bmatrix} 0.072 & 0.096 & 0 & 0 & -0.072 & -0.096 \\ 0.096 & 0.128 & 0 & 0 & -0.096 & -0.128 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -0.072 & -0.096 & 0 & 0 & 0.072 & 0.096 \\ -0.096 & -0.128 & 0 & 0 & 0.096 & 0.128 \end{bmatrix} \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{matrix} \quad \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 \end{matrix}$$

$$\mathbf{K} = AE\begin{bmatrix} 0.405 & 0.096 & -0.333 & 0 & -0.072 & -0.096 \\ 0.096 & 0.128 & 0 & 0 & -0.096 & -0.128 \\ -0.333 & 0 & 0.333 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -0.072 & -0.096 & 0 & 0 & 0.072 & 0.096 \\ -0.096 & -0.128 & 0 & 0 & 0.096 & 0.128 \end{bmatrix} \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{matrix} \quad \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 \end{matrix}$$

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Stiffness method of analysis: **trusses**

Application of the Stiffness Method for Truss Analysis

- Once the structure stiffness matrix is formed, the global force components **Q** acting on the truss can then be related to its global displacements **D** using: $\mathbf{Q} = \mathbf{KD}$
- This equation is referred to as the ***structure stiffness equation***.
- Since we have always assigned the ***lowest code numbers*** to identify the ***unconstrained degrees of freedom***, this will allow us now to partition this equation in the following form:

$$\begin{Bmatrix} \mathbf{Q}_k \\ \mathbf{Q}_u \end{Bmatrix} = \begin{bmatrix} \mathbf{K}_{11} & \mathbf{K}_{12} \\ \mathbf{K}_{21} & \mathbf{K}_{22} \end{bmatrix} \begin{Bmatrix} \mathbf{D}_u \\ \mathbf{D}_k \end{Bmatrix}$$

$\mathbf{Q}_k, \mathbf{D}_k$ ***known*** loads and displacements

$\mathbf{Q}_u, \mathbf{D}_u$ ***unknown*** loads and displacements

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Stiffness method of analysis: **trusses**

Application of the Stiffness Method for Truss Analysis

- Expanding the partitioned matrix yields:

$$\mathbf{Q}_k = \mathbf{K}_{11}\mathbf{D}_u + \mathbf{K}_{12}\mathbf{D}_k \qquad \mathbf{Q}_u = \mathbf{K}_{21}\mathbf{D}_u + \mathbf{K}_{22}\mathbf{D}_k$$

- Most often $\mathbf{D}_k = 0$ since the supports are not displaced.

$$\mathbf{Q}_k = \mathbf{K}_{11}\mathbf{D}_u$$

- Solving for $\mathbf{D}_u$ gives: $\mathbf{D}_u = \mathbf{K}_{11}^{-1}\mathbf{Q}_k$

- Here we can obtain a direct solution for all the unknown joint displacements: $\mathbf{Q}_u = \mathbf{K}_{21}\mathbf{D}_u$

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Stiffness method of analysis: **trusses**

Application of the Stiffness Method for Truss Analysis

- Member forces can be determined using: $q = k'\mathbf{TD}$

$$\begin{Bmatrix} q_N \\ q_F \end{Bmatrix} = \frac{AE}{L}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} \lambda_x & \lambda_y & 0 & 0 \\ 0 & 0 & \lambda_x & \lambda_y \end{bmatrix}\begin{Bmatrix} D_{Nx} \\ D_{Ny} \\ D_{Fx} \\ D_{Fy} \end{Bmatrix}$$

- Since $q_N = -q_F$ for equilibrium, only one of these forces has to be found.
- Here we will determine q_F, the one that exerts tension in the member:

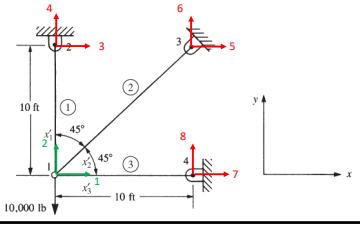
$$q_F = \frac{AE}{L}\begin{bmatrix} -\lambda_x & -\lambda_y & \lambda_x & \lambda_y \end{bmatrix}\begin{Bmatrix} D_{Nx} \\ D_{Ny} \\ D_{Fx} \\ D_{Fy} \end{Bmatrix}$$

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Stiffness method of analysis: **trusses**

Example 14.2 – Plane Truss Problem

- The plane truss shown below is composed of three bars subjected to a downward force of 10 *kips* at node 1. Assume the cross-sectional area $A = 2\ in^2$ and $E = 30 \times 10^6\ psi$ for all elements.
- Determine the *x* and *y* displacements at node 1 and the stresses in each element.

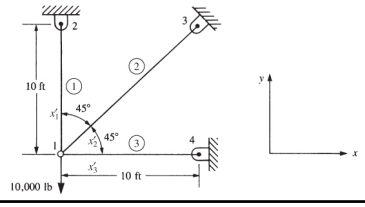


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Stiffness method of analysis: **trusses**

Example 14.2 – Plane Truss Problem

Element	Node 1	Node 2	θ_x	θ_y	λ_x	λ_y
1	1	2	90°	0°	0	1
2	1	3	45°	45°	0.707	0.707
3	1	4	0°	90°	1	0



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Stiffness method of analysis: **trusses**

Example 14.2 – Plane Truss Problem

- The global elemental stiffness matrix are:

$$\mathbf{k} = \frac{AE}{L}\begin{bmatrix} \lambda_x^2 & \lambda_x\lambda_y & -\lambda_x^2 & -\lambda_x\lambda_y \\ \lambda_x\lambda_y & \lambda_y^2 & -\lambda_x\lambda_y & -\lambda_y^2 \\ -\lambda_x^2 & -\lambda_x\lambda_y & \lambda_x^2 & \lambda_x\lambda_y \\ -\lambda_x\lambda_y & -\lambda_y^2 & \lambda_x\lambda_y & \lambda_y^2 \end{bmatrix}$$

element 1: $\lambda_x = 0 \quad \lambda_y = 1 \quad \Rightarrow \quad \mathbf{k}^{(1)} = \dfrac{(2in^2)(30\times10^6 psi)}{120in}$

$$\begin{array}{c} \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix} \end{array} \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} \; lb/in$$

element 2: $\lambda_x = \frac{\sqrt{2}}{2} \quad \lambda_y = \frac{\sqrt{2}}{2} \quad \Rightarrow \quad \mathbf{k}^{(2)} = \dfrac{(2in^2)(30\times10^6 psi)}{240\sqrt{2}in}$

$$\begin{array}{c} \begin{matrix} 1 & 2 & 5 & 6 \end{matrix} \\ \begin{bmatrix} 1 & 1 & -1 & -1 \\ 1 & 1 & -1 & -1 \\ -1 & -1 & 1 & 1 \\ -1 & -1 & 1 & 1 \end{bmatrix} \end{array} \begin{matrix} 1 \\ 2 \\ 5 \\ 6 \end{matrix} \; lb/in$$

element 3: $\lambda_x = 1 \quad = \lambda_y = 0 \quad \Rightarrow \quad \mathbf{k}^{(3)} = \dfrac{(2in^2)(30\times10^6 psi)}{120in}$

$$\begin{array}{c} \begin{matrix} 1 & 2 & 7 & 8 \end{matrix} \\ \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{array} \begin{matrix} 1 \\ 2 \\ 7 \\ 8 \end{matrix} \; lb/in$$

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Stiffness method of analysis: **trusses**

Example 14.2 – Plane Truss Problem

- The total global stiffness matrix is:

$$\mathbf{K} = 5\times10^5 \begin{bmatrix} 1.354 & 0.354 & 0 & 0 & -0.354 & -0.354 & -1 & 0 \\ 0.354 & 1.354 & 0 & -1 & -0.354 & -0.354 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 & 0 & 0 & 0 & 0 \\ -0.354 & -0.354 & 0 & 0 & 0.354 & 0.354 & 0 & 0 \\ -0.354 & -0.354 & 0 & 0 & 0.354 & 0.354 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} lb/in$$

element 1 element 2 element 3

- The total global force-displacement equations are:

$$\begin{Bmatrix} 0 \\ -10,000 \\ Q_3 \\ Q_4 \\ Q_5 \\ Q_6 \\ Q_7 \\ Q_8 \end{Bmatrix} = 5\times10^5 \begin{bmatrix} 1.354 & 0.354 & 0 & 0 & -0.354 & -0.354 & -1 & 0 \\ 0.354 & 1.354 & 0 & -1 & -0.354 & -0.354 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 & 0 & 0 & 0 & 0 \\ -0.354 & -0.354 & 0 & 0 & 0.354 & 0.354 & 0 & 0 \\ -0.354 & -0.354 & 0 & 0 & 0.354 & 0.354 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{Bmatrix} D_1 \\ D_2 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{Bmatrix}$$

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Stiffness method of analysis: **trusses**

Example 14.2 – Plane Truss Problem

- Applying the boundary conditions for the truss, the above equations reduce to:

$$\begin{Bmatrix} 0 \\ -10,000 \\ Q_3 \\ Q_4 \\ Q_5 \\ Q_6 \\ Q_7 \\ Q_8 \end{Bmatrix} = 5\times10^5 \begin{bmatrix} 1.354 & 0.354 & 0 & 0 & -0.354 & -0.354 & -1 & 0 \\ 0.354 & 1.354 & 0 & -1 & -0.354 & -0.354 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 & 0 & 0 & 0 & 0 \\ -0.354 & -0.354 & 0 & 0 & 0.354 & 0.354 & 0 & 0 \\ -0.354 & -0.354 & 0 & 0 & 0.354 & 0.354 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{Bmatrix} D_1 \\ D_2 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{Bmatrix}$$

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Stiffness method of analysis: **trusses**

Example 14.2 – Plane Truss Problem

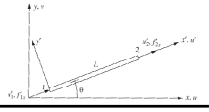
- Applying the boundary conditions for the truss, the above equations reduce to:

$$\begin{Bmatrix} 0 \\ -10,000 \end{Bmatrix} = 5\times10^5 \begin{bmatrix} 1.354 & 0.354 \\ 0.354 & 1.354 \end{bmatrix} \begin{Bmatrix} D_1 \\ D_2 \end{Bmatrix}$$

- Solving the equations gives: $D_1 = 0.414\times10^{-2}\,in$

$D_2 = -1.59\times10^{-2}\,in$

- Stress in an element is: $\sigma = \frac{E}{L}\left[-\lambda_x D_{Nx} - \lambda_y D_{Ny} + \lambda_x D_{Fx} + \lambda_y D_{Fy}\right]$



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Stiffness method of analysis: **trusses**

Example 14.2 – Plane Truss Problem

Element	Node 1	Node 2	θ_x	θ_y	λ_x	λ_y
1	1	2	90°	0°	0	1
2	1	3	45°	45°	0.707	0.707
3	1	4	0°	90°	1	0

$$\sigma = \frac{E}{L}\left[-\lambda_x D_{Nx} - \lambda_y D_{Ny} + \lambda_x D_{Fx} + \lambda_y D_{Fy}\right]$$

element 1 $\sigma^{(1)} = \frac{30\times10^6}{120}\left[-D_2\right] = 3,965\ psi$

element 2 $\sigma^{(2)} = -\frac{30\times10^6}{120}\left[(0.707)D_1 + (0.707)D_2\right] = 1,471\ psi$

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Stiffness method of analysis: **trusses**

Example 14.2 – Plane Truss Problem

Element	Node 1	Node 2	θ_x	θ_y	λ_x	λ_y
1	1	2	90°	0°	0	1
2	1	3	45°	45°	0.707	0.707
3	1	4	0°	90°	1	0

$$\sigma = \frac{E}{L}\left[-\lambda_x D_{Nx} - \lambda_y D_{Ny} + \lambda_x D_{Fx} + \lambda_y D_{Fy}\right]$$

element 3 $\sigma^{(3)} = \frac{30\times10^6}{120}\left[-D_1\right] = -1,035\ psi$

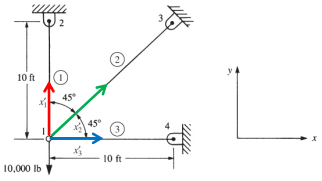
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Stiffness method of analysis: **trusses**

Example 14.2 – Plane Truss Problem

- Let's check equilibrium at node 1:

$F = \sigma A$



$\sum F_x = F^{(2)}\cos(45°) + F^{(3)}$

$\sum F_y = F^{(2)}\sin(45°) + F^{(1)} - 10,000\,lb$

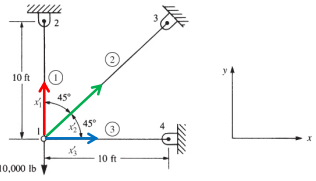
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Stiffness method of analysis: **trusses**

Example 14.2 – Plane Truss Problem

- Let's check equilibrium at node 1:

$F = \sigma A$



$\sum F_x = (1,471\ psi)(2\,in^2)(0.707) - (1,035\ psi)(2\,in^2) = 0$

$\sum F_y = (3,965\ psi)(2\,in^2) + (1,471\ psi)(2\,in^2)(0.707) - 10,000 = 0$

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Stiffness method of analysis: **trusses**

Let's work on some problems

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Stiffness method of analysis: **trusses**

Any questions?



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